

## Appendix

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Our appendix is organized into three sections:

- Appendix [A](#) provides additional methodological details and derivations, including second-order Tweedie uncertainty, Hutchinson-JVP diagonal estimation, uncertainty calibration, uncertainty-calibrated posterior anchoring, group-relative refinement, and the refined clean prediction.
- Appendix [B](#) presents the complete pseudocode of the proposed training-free inference procedure, including uncertainty estimation, posterior anchoring, group-relative refinement, and the DDIM reverse update.
- Appendix [C](#) provides the experimental configurations, including baseline settings and an analysis of the calibration scale  $\gamma$ .

## A Additional Methodological Details

This supplementary material provides additional derivations and implementation details for the proposed framework. We first derive the second-order Tweedie uncertainty and its Hutchinson-JVP approximation. We then provide a detailed derivation of the uncertainty-calibrated posterior anchor and its measurement-residual contraction property. Finally, we describe the uncertainty-aware group-relative refinement and DDIM reverse update in greater detail.

### A.1 Detailed Derivation of Second-Order Tweedie Uncertainty

We consider the variance-preserving forward diffusion process

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (31)$$

Define

$$\tilde{\mathbf{x}}_t = \frac{\mathbf{x}_t}{\sqrt{\bar{\alpha}_t}}, \quad \tau_t^2 = \frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t}, \quad (32)$$

such that

$$\tilde{\mathbf{x}}_t = \mathbf{x}_0 + \tau_t \boldsymbol{\epsilon}. \quad (33)$$

Let

$$\boldsymbol{\mu}_t = \mathbb{E}[\mathbf{x}_0 | \tilde{\mathbf{x}}_t] \quad (34)$$

denote the conditional clean-image mean.

*Conditional covariance identity.* For the Gaussian perturbation model in Eq. (33), the posterior density satisfies

$$p(\mathbf{x}_0 | \tilde{\mathbf{x}}_t) \propto p(\mathbf{x}_0) \exp \left( -\frac{\|\tilde{\mathbf{x}}_t - \mathbf{x}_0\|_2^2}{2\tau_t^2} \right). \quad (35)$$

For coordinates  $(i, j)$ ,

$$\frac{\partial p(\mathbf{x}_0 | \tilde{\mathbf{x}}_t)}{\partial \tilde{x}_{t,j}} = \frac{x_{0,j} - \mu_{t,j}}{\tau_t^2} p(\mathbf{x}_0 | \tilde{\mathbf{x}}_t). \quad (36)$$

Differentiating the conditional mean gives

$$\frac{\partial \mu_{t,i}}{\partial \tilde{x}_{t,j}} = \int x_{0,i} \frac{\partial p(\mathbf{x}_0 | \tilde{\mathbf{x}}_t)}{\partial \tilde{x}_{t,j}} d\mathbf{x}_0 \quad (37)$$

$$= \frac{1}{\tau_t^2} \mathbb{E}[x_{0,i}(x_{0,j} - \mu_{t,j}) | \tilde{\mathbf{x}}_t] \quad (38)$$

$$= \frac{1}{\tau_t^2} \text{Cov}[x_{0,i}, x_{0,j} | \tilde{\mathbf{x}}_t]. \quad (39)$$

Collecting all entries yields

$$\text{Cov}[\mathbf{x}_0 | \tilde{\mathbf{x}}_t] = \tau_t^2 \nabla_{\tilde{\mathbf{x}}_t} \boldsymbol{\mu}_t. \quad (40)$$

*Connection to the denoiser.* For a noise-prediction diffusion model, the Tweedie clean prediction is

$$\hat{\mathbf{x}}_{0|t} = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}}. \quad (41)$$

Since  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \tilde{\mathbf{x}}_t$ , we can equivalently write

$$\hat{\mathbf{x}}_{0|t} = \tilde{\mathbf{x}}_t - \tau_t \epsilon_\theta(\sqrt{\bar{\alpha}_t} \tilde{\mathbf{x}}_t, t). \quad (42)$$

Let

$$\mathbf{J}_{\epsilon, t} = \frac{\partial \epsilon_\theta(\mathbf{x}_t, t)}{\partial \mathbf{x}_t}. \quad (43)$$

Applying the chain rule gives

$$\nabla_{\tilde{\mathbf{x}}_t} \hat{\mathbf{x}}_{0|t} = \mathbf{I} - \tau_t \sqrt{\bar{\alpha}_t} \mathbf{J}_{\epsilon, t} \quad (44)$$

$$= \mathbf{I} - \sqrt{1 - \bar{\alpha}_t} \mathbf{J}_{\epsilon, t}. \quad (45)$$

Substituting Eq. (45) into Eq. (40) yields

$$\Sigma_{0|t} = \frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t} (\mathbf{I} - \sqrt{1 - \bar{\alpha}_t} \mathbf{J}_{\epsilon, t}). \quad (46)$$

For an exact conditional denoiser, Eq. (46) corresponds to the conditional covariance. With a learned denoiser, we instead use it as a local model-based approximation of the clean-image uncertainty.

## A.2 Hutchinson-JVP Diagonal Estimation

Explicit construction of  $\mathbf{J}_{\epsilon, t}$  is infeasible for high-dimensional images. Since the proposed method requires only spatially varying marginal uncertainty, we estimate its diagonal using Hutchinson’s estimator.

Let  $\mathbf{v} \in \{-1, +1\}^d$  be a Rademacher random vector with independent entries. For the  $i$ -th component,

$$[\mathbf{v} \odot (\mathbf{J}\mathbf{v})]_i = v_i \sum_j J_{ij} v_j = \sum_j J_{ij} v_i v_j. \quad (47)$$

Since

$$\mathbb{E}[v_i v_j] = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad (48)$$

we obtain

$$\mathbb{E}_{\mathbf{v}} [\mathbf{v} \odot (\mathbf{J}\mathbf{v})] = \text{diag}(\mathbf{J}). \quad (49)$$

Using  $P$  independent Rademacher probes,

$$\widehat{\text{diag}}(\mathbf{J}_{\epsilon, t}) = \frac{1}{P} \sum_{p=1}^P \mathbf{v}_p \odot (\mathbf{J}_{\epsilon, t} \mathbf{v}_p), \quad \mathbf{v}_p \in \{-1, +1\}^d. \quad (50)$$

This provides an unbiased stochastic estimate of the Jacobian diagonal. Each product  $\mathbf{J}_{\epsilon,t}\mathbf{v}_p$  is evaluated through a Jacobian-vector product (JVP), avoiding explicit construction of the denoiser Jacobian.

The corresponding raw diagonal uncertainty estimate is

$$\hat{\mathbf{U}}_t = \frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t} \left( \mathbf{1} - \sqrt{1 - \bar{\alpha}_t} \widehat{\text{diag}}(\mathbf{J}_{\epsilon,t}) \right). \quad (51)$$

### A.3 Uncertainty Calibration

Because the learned denoiser and stochastic diagonal estimator can produce negative or unstable values, we transform the raw estimate  $\hat{\mathbf{U}}_t$  into a bounded uncertainty map:

$$\mathbf{M}_t = \tanh \left( \gamma \text{softplus}(\hat{\mathbf{U}}_t) \right), \quad \mathbf{M}_t \in [0, 1]. \quad (52)$$

The softplus operation ensures non-negativity, while the hyperbolic tangent bounds the resulting uncertainty magnitude.

Consistent with the main formulation, the covariance used for observation conditioning is

$$\mathbf{C}_t = \text{Diag}(\mathbf{M}_t). \quad (53)$$

Here,  $\mathbf{M}_t$  is interpreted as a bounded uncertainty surrogate rather than the exact posterior variance. The spatially varying covariance  $\mathbf{C}_t$  subsequently modulates the observation correction according to the estimated reliability of the diffusion prediction.

### A.4 Derivation of Uncertainty-Calibrated Posterior Anchoring

At reverse timestep  $t$ , we approximate the clean-image belief as

$$p(\mathbf{x}_0|\mathbf{x}_t) \approx \mathcal{N}(\hat{\mathbf{x}}_{0|t}, \mathbf{C}_t), \quad (54)$$

and consider the linear Gaussian observation model

$$\mathbf{y} = \mathbf{H}\mathbf{x}_0 + \mathbf{n}, \quad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{I}). \quad (55)$$

**Conditional Posterior Mean** Under the local Gaussian approximation, the joint distribution of  $\mathbf{x}_0$  and  $\mathbf{y}$ , conditioned on  $\mathbf{x}_t$ , is

$$\begin{bmatrix} \mathbf{x}_0 \\ \mathbf{y} \end{bmatrix} \sim \mathcal{N} \left( \begin{bmatrix} \hat{\mathbf{x}}_{0|t} \\ \mathbf{H}\hat{\mathbf{x}}_{0|t} \end{bmatrix}, \begin{bmatrix} \mathbf{C}_t & \mathbf{C}_t \mathbf{H}^\top \\ \mathbf{H} \mathbf{C}_t & \mathbf{H} \mathbf{C}_t \mathbf{H}^\top + \sigma_y^2 \mathbf{I} \end{bmatrix} \right). \quad (56)$$

For a jointly Gaussian pair,

$$\mathbb{E}[\mathbf{x}_0|\mathbf{x}_t, \mathbf{y}] = \hat{\mathbf{x}}_{0|t} + \text{Cov}(\mathbf{x}_0, \mathbf{y}) \text{Cov}(\mathbf{y})^{-1} (\mathbf{y} - \mathbb{E}[\mathbf{y}]). \quad (57)$$

From Eq. (56),

$$\text{Cov}(\mathbf{x}_0, \mathbf{y}) = \mathbf{C}_t \mathbf{H}^\top, \quad \text{Cov}(\mathbf{y}) = \mathbf{H} \mathbf{C}_t \mathbf{H}^\top + \sigma_y^2 \mathbf{I}, \quad \mathbb{E}[\mathbf{y}] = \mathbf{H} \hat{\mathbf{x}}_{0|t}. \quad (58)$$

Therefore,

$$\mathbf{x}_{0,t}^{\text{anc}} = \hat{\mathbf{x}}_{0|t} + \mathbf{C}_t \mathbf{H}^\top (\mathbf{H} \mathbf{C}_t \mathbf{H}^\top + \sigma_y^2 \mathbf{I})^{-1} (\mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_{0|t}). \quad (59)$$

Thus,  $\mathbf{x}_{0,t}^{\text{anc}}$  is the conditional Gaussian mean under the local belief in Eq. (54).

**Measurement-Residual Contraction** We next show that posterior anchoring does not increase the measurement residual relative to the Tweedie prediction. Define

$$\mathbf{r}_t = \mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_{0|t}, \quad \mathbf{Q}_t = \mathbf{H} \mathbf{C}_t \mathbf{H}^\top. \quad (60)$$

Since  $\mathbf{C}_t \succeq 0$ , we have  $\mathbf{Q}_t \succeq 0$  because, for any vector  $\mathbf{u}$ ,

$$\mathbf{u}^\top \mathbf{Q}_t \mathbf{u} = (\mathbf{H}^\top \mathbf{u})^\top \mathbf{C}_t (\mathbf{H}^\top \mathbf{u}) \geq 0. \quad (61)$$

Substituting Eq. (59) gives

$$\mathbf{y} - \mathbf{H} \mathbf{x}_{0,t}^{\text{anc}} = \mathbf{r}_t - \mathbf{Q}_t (\mathbf{Q}_t + \sigma_y^2 \mathbf{I})^{-1} \mathbf{r}_t \quad (62)$$

$$= \left[ \mathbf{I} - \mathbf{Q}_t (\mathbf{Q}_t + \sigma_y^2 \mathbf{I})^{-1} \right] \mathbf{r}_t. \quad (63)$$

Using

$$\mathbf{I} - \mathbf{Q}_t (\mathbf{Q}_t + \sigma_y^2 \mathbf{I})^{-1} = \sigma_y^2 (\mathbf{Q}_t + \sigma_y^2 \mathbf{I})^{-1}, \quad (64)$$

we obtain

$$\mathbf{y} - \mathbf{H} \mathbf{x}_{0,t}^{\text{anc}} = \sigma_y^2 (\mathbf{Q}_t + \sigma_y^2 \mathbf{I})^{-1} \mathbf{r}_t. \quad (65)$$

Let  $q_j \geq 0$  be an eigenvalue of  $\mathbf{Q}_t$ . The corresponding eigenvalue of the residual transformation is

$$\frac{\sigma_y^2}{q_j + \sigma_y^2} \leq 1. \quad (66)$$

Hence,

$$\|\mathbf{y} - \mathbf{H} \mathbf{x}_{0,t}^{\text{anc}}\|_2 \leq \|\mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_{0|t}\|_2. \quad (67)$$

Therefore, posterior anchoring incorporates observation information without increasing the measurement residual relative to the Tweedie prediction. It provides a measurement-informed reference for subsequent group-relative refinement.

**Matrix-Free Posterior Update** Explicit construction or inversion of  $\mathbf{H} \mathbf{C}_t \mathbf{H}^\top + \sigma_y^2 \mathbf{I}$  is unnecessary. Instead, we solve

$$(\mathbf{H} \mathbf{C}_t \mathbf{H}^\top + \sigma_y^2 \mathbf{I}) \mathbf{v}_t = \mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_{0|t} \quad (68)$$

using conjugate gradient, and compute

$$\mathbf{x}_{0,t}^{\text{anc}} = \hat{\mathbf{x}}_{0|t} + \mathbf{C}_t \mathbf{H}^\top \mathbf{v}_t. \quad (69)$$

For  $\sigma_y^2 > 0$ , the system matrix is symmetric positive definite. For any  $\mathbf{v} \neq \mathbf{0}$ ,

$$\mathbf{v}^\top (\mathbf{H}\mathbf{C}_t\mathbf{H}^\top + \sigma_y^2\mathbf{I}) \mathbf{v} \quad (70)$$

$$= (\mathbf{H}^\top \mathbf{v})^\top \mathbf{C}_t (\mathbf{H}^\top \mathbf{v}) + \sigma_y^2 \|\mathbf{v}\|_2^2 > 0. \quad (71)$$

Thus, only applications of  $\mathbf{H}$  and its adjoint  $\mathbf{H}^\top$  are required. When an explicit adjoint is unavailable, for a linear forward operator  $\mathbf{H}(\mathbf{x})$  we evaluate

$$\mathbf{H}^\top \mathbf{v} = \nabla_{\mathbf{x}} \langle \mathbf{H}(\mathbf{x}), \mathbf{v} \rangle. \quad (72)$$

### A.5 Additional Details of Group-Relative Refinement

Posterior anchoring incorporates information supported by the observation, but information-losing operators may leave ambiguity in directions that are weakly constrained by the measurement. We therefore perform uncertainty-aware group-relative refinement around the posterior anchor in a low-dimensional search space.

**Uncertainty-Guided Proposal Sampling** Let

$$\mathbf{m}_t \in \mathbb{R}^{C \times L \times L}, \quad L \ll H, W, \quad (73)$$

denote the search mean. At each active reverse step, we initialize

$$\mathbf{m}_t = \mathbf{0}. \quad (74)$$

The uncertainty map is pooled to the same spatial resolution,

$$\mathbf{U}_t^z = \text{Pool}_{L \times L}(\mathbf{M}_t), \quad \boldsymbol{\sigma}_t^z = \sqrt{\mathbf{U}_t^z + \varepsilon}. \quad (75)$$

Candidate proposals are sampled as

$$\mathbf{z}_t^{(i)} = \mathbf{m}_t + \boldsymbol{\sigma}_t^z \odot \boldsymbol{\epsilon}^{(i)}, \quad \boldsymbol{\epsilon}^{(i)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (76)$$

Thus, regions with higher estimated uncertainty are explored more broadly, whereas more reliable regions remain closer to the current search mean. Uncertainty controls the diversity of candidate exploration rather than directly specifying the refinement update.

**Antithetic Candidate Groups** For an even group size  $N$ , we draw  $N/2$  independent Gaussian perturbations and use their negatives to construct

$$\{\boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_{N/2}, -\boldsymbol{\epsilon}_1, \dots, -\boldsymbol{\epsilon}_{N/2}\}. \quad (77)$$

Consequently,

$$\frac{1}{N} \sum_{i=1}^N \boldsymbol{\epsilon}^{(i)} = \mathbf{0}. \quad (78)$$

This construction provides symmetric exploration around  $\mathbf{m}_t$  and reduces asymmetry caused by finite candidate sampling.

**Candidate Construction** Each low-dimensional proposal is mapped to image space through  $\mathcal{B}$ , producing

$$\mathbf{x}_{0,t}^{(i)} = \mathbf{x}_{0,t}^{\text{anc}} + \lambda_t \mathcal{B}(\mathbf{z}_t^{(i)}). \quad (79)$$

The refinement strength follows the linear schedule

$$p_t = \text{clip} \left( \frac{t_{\text{start}} - t}{t_{\text{start}} - t_{\text{end}}}, 0, 1 \right), \quad \lambda_t = \lambda_{\text{max}} p_t. \quad (80)$$

Hence,  $\lambda_t = 0$  at  $t_{\text{start}}$  and increases progressively toward  $\lambda_{\text{max}}$  as the reverse trajectory approaches  $t_{\text{end}}$ .

**Measurement–Perceptual Reward** Each candidate is evaluated using two complementary scalar signals: measurement fidelity and no-reference image quality,

$$r_i = r_{\text{meas}}(\mathbf{x}_{0,t}^{(i)}) + Q(\mathbf{x}_{0,t}^{(i)}), \quad r_{\text{meas}}(\mathbf{x}) = -\|\mathbf{H}\mathbf{x} - \mathbf{y}\|_2^2. \quad (81)$$

The two terms use unit coefficients. The measurement reward assigns higher values to candidates with smaller measurement error, while  $Q$  provides a no-reference perceptual ranking signal without access to the unknown ground-truth image. Both terms are used only as scalar evaluations during group-relative refinement.

**Group-Relative Advantages** Candidate rewards are standardized within each group:

$$a_i = \frac{r_i - \mu_r}{\sigma_r + \varepsilon}, \quad \mu_r = \frac{1}{N} \sum_{i=1}^N r_i, \quad \sigma_r^2 = \frac{1}{N} \sum_{i=1}^N (r_i - \mu_r)^2. \quad (82)$$

Positive advantages correspond to candidates performing above the group average, while negative advantages correspond to candidates performing below it. By construction,

$$\sum_{i=1}^N a_i = 0. \quad (83)$$

**Advantage-Weighted Proposal Aggregation** The group-relative update aggregates proposals according to their relative advantages:

$$\mathbf{g}_t = \frac{1}{N} \sum_{i=1}^N a_i \mathbf{z}_t^{(i)}. \quad (84)$$

Using Eq. (83),

$$\mathbf{g}_t = \frac{1}{N} \sum_{i=1}^N a_i \left( \mathbf{z}_t^{(i)} - \mathbf{m}_t \right) \quad (85)$$

$$= \frac{1}{N} \sum_{i=1}^N a_i \left( \boldsymbol{\sigma}_t^z \odot \boldsymbol{\epsilon}^{(i)} \right). \quad (86)$$

Thus, the search mean itself does not introduce a systematic shift in the aggregation. Instead, the update depends on how the group-relative rewards correlate with the sampled candidate variations. Equation (86) also makes the complementary roles explicit:  $\boldsymbol{\sigma}_t^z$  controls the available exploration, whereas  $a_i$  determines which sampled variations are favored.

The search mean is updated as

$$\mathbf{m}_t \leftarrow \mathbf{m}_t + \eta_{\text{GR}} \mathbf{g}_t. \quad (87)$$

This procedure is repeated for  $J$  inner refinement iterations.

## A.6 Refined Clean Prediction

After  $J$  inner refinement iterations, only the optimized search mean is retained and mapped back to image space:

$$\mathbf{x}_{0,t}^* = \mathbf{x}_{0,t}^{\text{anc}} + \lambda_t \mathcal{B}(\mathbf{m}_t). \quad (88)$$

Individual candidate proposals are discarded after refinement and do not appear directly in the final prediction.

The complete refined clean prediction can be decomposed as

$$\mathbf{x}_{0,t}^* = \underbrace{\hat{\mathbf{x}}_{0|t}}_{\text{diffusion prediction}} + \underbrace{\mathbf{K}_t (\mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_{0|t})}_{\text{posterior anchoring}} + \underbrace{\lambda_t \mathcal{B}(\mathbf{m}_t)}_{\text{group-relative refinement}}, \quad (89)$$

where

$$\mathbf{K}_t = \mathbf{C}_t \mathbf{H}^\top (\mathbf{H} \mathbf{C}_t \mathbf{H}^\top + \sigma_y^2 \mathbf{I})^{-1}. \quad (90)$$

This decomposition highlights the complementary roles of the two stages. Posterior anchoring incorporates observation information, while group-relative refinement explores candidate variations that remain ambiguous after conditioning. The same uncertainty estimate is used in both stages: it calibrates the posterior correction through  $\mathbf{C}_t$  and controls candidate exploration through  $\boldsymbol{\sigma}_t^z$ .

## B Pseudocode

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**Algorithm 1** Proposed Training-Free Inference Procedure
 

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**Require:** Observation  $\mathbf{y}$ , operator  $\mathbf{H}$ , pretrained denoiser  $\epsilon_\theta$ ,  $P$ ,  $N$ ,  $J$ ,  $\gamma$ ,  $\sigma_y^2$ ,  $\eta_{\text{GR}}$ ,  $\lambda_{\text{max}}$ ,  $t_{\text{start}}$ ,  $t_{\text{end}}$

- 1: Initialize  $\mathbf{x}_T$
- 2: **for** each reverse timestep  $t$  **do**
- 3:   Compute Tweedie prediction  $\hat{\mathbf{x}}_{0|t}$
- 4:   **if**  $t_{\text{end}} \leq t \leq t_{\text{start}}$  **then**
- 5:     Estimate  $\text{diag}(\mathbf{J}_{\epsilon,t})$  using  $P$  Hutchinson-JVP probes
- 6:     Compute raw uncertainty  $\hat{\mathbf{U}}_t$  and  $\mathbf{M}_t = \tanh(\gamma \text{softplus}(\hat{\mathbf{U}}_t))$
- 7:     Set  $\mathbf{C}_t = \text{Diag}(\mathbf{M}_t)$
- 8:     Solve  $(\mathbf{H}\mathbf{C}_t\mathbf{H}^\top + \sigma_y^2\mathbf{I})\mathbf{v}_t = \mathbf{y} - \mathbf{H}\hat{\mathbf{x}}_{0|t}$  using conjugate gradient
- 9:     Compute  $\mathbf{x}_{0,t}^{\text{anc}} = \hat{\mathbf{x}}_{0|t} + \mathbf{C}_t\mathbf{H}^\top\mathbf{v}_t$
- 10:    Initialize  $\mathbf{m}_t \leftarrow \mathbf{0}$
- 11:    Compute  $\sigma_t^z = \sqrt{\text{Pool}_{L \times L}(\mathbf{M}_t) + \varepsilon}$
- 12:    Set  $p_t = \text{clip}((t_{\text{start}} - t)/(t_{\text{start}} - t_{\text{end}}), 0, 1)$  and  $\lambda_t = \lambda_{\text{max}}p_t$
- 13:    **for**  $j = 1, \dots, J$  **do**
- 14:     Draw  $N/2$  Gaussian perturbations and form an antithetic group of size  $N$
- 15:     **for**  $i = 1, \dots, N$  **do**
- 16:        $\mathbf{z}_t^{(i)} \leftarrow \mathbf{m}_t + \sigma_t^z \odot \epsilon^{(i)}$
- 17:        $\mathbf{x}_{0,t}^{(i)} \leftarrow \mathbf{x}_{0,t}^{\text{anc}} + \lambda_t \mathcal{B}(\mathbf{z}_t^{(i)})$
- 18:        $r_i \leftarrow -\|\mathbf{H}\mathbf{x}_{0,t}^{(i)} - \mathbf{y}\|_2^2 + Q(\mathbf{x}_{0,t}^{(i)})$
- 19:     **end for**
- 20:     Standardize  $\{r_i\}_{i=1}^N$  to obtain advantages  $\{a_i\}_{i=1}^N$
- 21:      $\mathbf{g}_t \leftarrow \frac{1}{N} \sum_{i=1}^N a_i(\sigma_t^z \odot \epsilon^{(i)})$
- 22:      $\mathbf{m}_t \leftarrow \mathbf{m}_t + \eta_{\text{GR}}\mathbf{g}_t$
- 23:    **end for**
- 24:     $\mathbf{x}_{0,t}^* \leftarrow \mathbf{x}_{0,t}^{\text{anc}} + \lambda_t \mathcal{B}(\mathbf{m}_t)$
- 25:    **else**
- 26:      $\mathbf{x}_{0,t}^* \leftarrow \hat{\mathbf{x}}_{0|t}$
- 27:    **end if**
- 28:     $\mathbf{x}_{t-1} \leftarrow \text{DDIM}(\mathbf{x}_t, \mathbf{x}_{0,t}^*)$
- 29: **end for**
- 30: **return**  $\mathbf{x}_0$

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## C Baseline Configurations

### C.1 Baselines

Table 4 summarizes the input resolution and baseline-specific guidance hyperparameters used for each inverse-problem task. We report the actual hyperparameter values used in our experiments without introducing method-specific notation

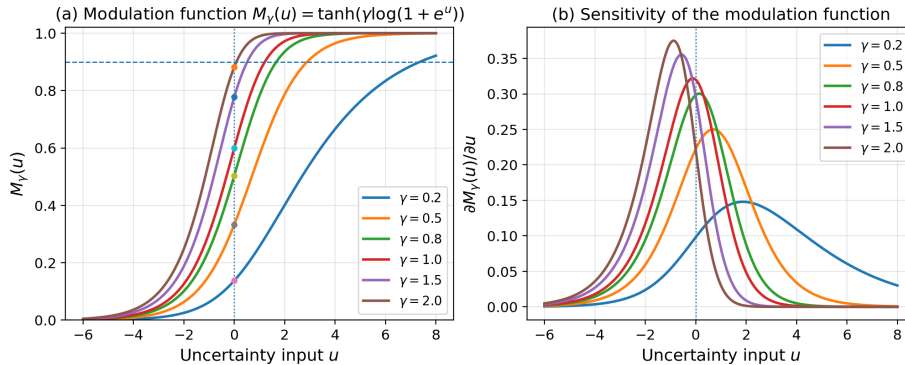


Fig. 5: Effect of the calibration scale  $\gamma$  on the uncertainty transformation  $\mathbf{M}_t = \tanh(\gamma \text{softplus}(\hat{\mathbf{U}}_t))$ . Larger  $\gamma$  produces a steeper response but causes earlier saturation toward 1, reducing discrimination between different uncertainty levels.

unless explicitly required. All baselines use the same  $256 \times 256$  input resolution and are evaluated under the same degradation and sampling protocol.

Table 4: Input resolution and baseline-specific guidance hyperparameters used for Gaussian deblurring and super-resolution.

| Task                | Input Size       | DPS | LGD | MPGD | FreeDoM | UGD |
|---------------------|------------------|-----|-----|------|---------|-----|
| Gaussian Deblurring | $256 \times 256$ | 16  | 16  | 16   | 16      | 16  |
| Super-Resolution    | $256 \times 256$ | 16  | 16  | 16   | 16      | 8   |

## C.2 Effect of the Calibration Scale $\gamma$

As illustrated in Fig. 5, the calibration scale  $\gamma$  controls the sensitivity of the bounded uncertainty map  $\mathbf{M}_t$  to the raw second-order estimate  $\hat{\mathbf{U}}_t$ . Increasing  $\gamma$  produces a steeper transformation, assigning larger uncertainty values to the same input. However, excessively large values also cause  $\mathbf{M}_t$  to saturate more rapidly toward 1, compressing the differences between regions with different uncertainty levels.

This saturation affects both stages of the framework. In posterior anchoring,  $\mathbf{C}_t = \text{Diag}(\mathbf{M}_t)$  becomes more uniform, reducing the spatial adaptivity of the observation correction. In group-relative refinement, it also makes the exploration scale more uniform, weakening the distinction between reliable and uncertain regions. Conversely, very small  $\gamma$  reduces the dynamic range of the calibrated uncertainty. We therefore use a moderate value of  $\gamma = 0.2$ .